Class	$[0;10[$	$[10;20[$	$[20;30[$
Count	4	9	7

What is the total size, modal class and frequency of $[10;20[$?

Rep. 20 ; $[10;20[$; $\frac{9}{20} = 45\,\%$.

5. Red thread · Problem [IVD] One bus passes every 15 minutes, another every 20 minutes, and they cross at 8 a.m. What time do they meet again?

Rep. PPCM $(15\,;\,20) = 60$: à 9 h.

Form 17 Mon. 30 Nov. 2026 Absolute value and distance between two reals

1. Warming · Flash [REP] What is the distance between $-$Three and five on the graduated right?

Rep. $|5-(-3)| = 8$.

2. Heart · Flash [CAL] Calculate $|-7|$, $|3-8|$ and $|\pi-4|$.

Rep. $7\,;\,5\,;\,|\pi-4| = 4-\pi$ since $\pi < 4$.

3. Heart + · Method [EQ] Solve $|x-3| = 2$.

Rep. The real at distance 2 of 3 : $x = 1$ or $x = 5$.

Geometrical interpretation rather than case by case.

4. Back · Formula [IMG] $f(x) = x^2 - 3$. Give the image of -2 and preimage of 6.

Rep. $f(-2) = 1\,;\,x^2 - 3 = 6 \Leftrightarrow x^2 = 9$: Preimages -3 and 3.

5. Red thread · Formula → table [REG] $f(x) = -x + 3$. Complete: $f(-1), f(0), f(2)$.

Rep. $4\,;\,3\,;\,1$.

Form 18 Mar. 1 1st Dec. 2026 Inequation $|x-a| \leqslant$ r and associated interval

1. Warming · Flash [EQ] Solve 2 $x - 3 \leqslant 5$.

Rep. $2x \leqslant 8$, $x \leqslant 4$: $S =]-\infty;4]$.

2. Heart · Method [EQ] Solve $|x-2| \leqslant 3$ and give S in the form of interval.

Rep. The real at a distance of up to 3 of 2: $S = [-1;5]$.

Control: $|-1-2| = 3$ *et* $|5-2| = 3$.

3. Heart + · Translate [NUM] Translated by inequality with absolute value: $x \in [-1;5]$ then $x \in]4;10[$.

Rep. $|x-2| \leqslant 3$ (centre 2, radius 3); $|x-7| < 3$ (centre 7, radius 3).

4. Back · Problem [CMP] Lot A : 6 yogurts for 4.20 €. Lot B : 10 yogurt for 6,80 €. Which is the most advantageous?

Rep. A : 0,70 € Unit; B: 0.68 € : B.

5. Red thread · Flash [GEO] A disc has a radius of 5 cm. Give its area and perimeter (accurate values).

Rep. Area $= \pi \times 5^2 = 25\pi$ cm^2 ; perimeter $= 2\pi \times 5 = 10\pi$ Inch.

Functions: representations and reference functions

Form 19 3 Dec. 2026 Function with real values; definition set

1. Warming · Flash [EQ] Solve $x - 4 = 0$ and 2 $x + 6 = 0$.

Rep. $x = 4\,;\,x = -3$.

2. Heart · Method [MOD] Set of definitions of $f(x) = \dfrac{1}{x-4}$?

Rep. Excludes the value that cancels the denominator: $\mathbb{R} \setminus \{4\} =]-\infty;4[\cup]4;+\infty[$.

3. Heart + · MCQ [MOD] The definition set of $h(x) = \dfrac{1}{x^2+1}$ is:

A. $\mathbb{R} \setminus \{1\}$ **B.** $\mathbb{R} \setminus \{-1\,;\,1\}$ **C.** $\mathbb{R}$ **D.** $\mathbb{R}^*$

Rep. C. $\mathbb{R}$

$x^2+1 \geqslant 1 > 0$: the denominator never cancels.

4. Back · Table [PNP] Is this table proportionate?

x	2	5	8
y	6	15	24

Rep. Yes: $\frac{y}{x} = 3$ for each column.

5. Red thread · Code reading [ALG] What's sent back $f(4)$?

```
def f(x):
    return 3*x - 2
```

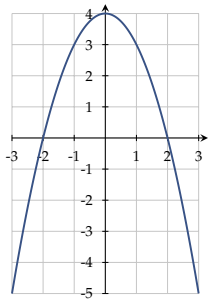
Rep. 10.

Form 20 Mon. 7 Dec 2026 Equation curve y $= f(x)$; images and preimages

1. Warming · Flash [IMG] $f(x) = -x^2 + 4$. Calculate $f(0)$, $f(-2)$ and $f(3)$.

Rep. 4 ; 0 ; -5.

2. Heart · Graph [IMG] Curve of f :



Read $f(1)$ and the preimage of 0.

Rep. $f(1) = 3$; preimage of 0 : -2 and 2.

Expression used for the route: f $(x) = -x^2 + 4$.

3. Heart + · Reflection [IMG] For this function, the equation $f(x) =$ Does 5 have any solutions?

Rep. No: $f(x) = 4 - x^2 \leqslant 4 < 5$ for all x ; the curve does not rise up to $y = 5$.

4. Back · Flash [PRB] Urn: 5 red balls, 3 blue, 2 green. Likely to pull a blue? A no Red?

Rep. $\frac{3}{10}$; $\frac{5}{10} = \frac{1}{2}$.

5. Red thread · Method [EQ] Solve in $\mathbb{R} : 4 - 3x = 10$.

Rep. $x = -2$

Trap: divide by -3 without changing the result sign.

Form 21 Mar. 8 Dec. 2026 Reference functions: absolute, square, inverse

1. Warming · Flash [CAL] Calculate $|-2{,}5|$, $(-3)^2$ and $\frac{1}{4}$.

Rep. $2{,}5$; 9 ; $-0{,}25$.

2. Heart · Table [REG] Complete for $x = -2$; -1 ; 0 ; 1 ; 2 $x \mapsto |x|$, $x \mapsto x^2$ and $x \mapsto \frac{1}{x}$.

Rep. $|x|$: 2 ; 1 ; 0 ; 1 ; 2 ; x^2 : 4 ; 1 ; 0 ; 1 ; 4 ; $\frac{1}{x}$: $-\frac{1}{2}$; -1 ; not defined ; 1 ; $1\frac{1}{2}$.

3. Heart + · MCQ [REG] What is the reference curve? $(-1;1)$, $(0;0)$, $(1;1)$ and $(2;4)$?

A. the right $y = x$ **B.** the other way round **C.** value absolute **D.** the square one

Rep. D. the square

The absolute value would pass through $(2;2)$; The inverse is not defined in 0 ; the right y = x passes through $(-1;-1)$.

4. Back · Flash [PRO] An object is enlarged with the ratio 1.5. What happens to a length of 8 cm? By What number is it multiplied?

Rep. 12 cm; the area is multiplied by 1.5 cm^2 = 2,25.

5. Red thread · Prove [LIT] Conjecture: "the sum of two consecutive integers is odd."

Rep. $n + (n+1) = 2n + 1$ of form 2 $k + 1$ with $k = n$: odd.

Form 22 Thu. 10 Dec. 2026 Square and cube root; sign of reference functions

1. Warming · Flash [CAL] Calculate $\overline{49}$, $\sqrt{0.25}$ and $(-2)^3$.

Rep. 7 ; $0{,}5$; -8.

2. Heart · Method [MOD] Study the sign of $f(x) = -2x + 6$.

Rep. $-2x + 6 = 0 \Leftrightarrow x = 3$; $f(x) > 0$ for $x < 3$, $f(x) < 0$ for $x > 3$ (negative slope).

3. Heart + · True/false [MOD] True or false? Justify. For all real x, $\sqrt{x^2} = x$.

Rep. Wrong. Counterexample: $x = -3$: $\sqrt{9} = 3 \neq -3$. In fact $\sqrt{x^2} = |x|$.

4. Back · Flash [Sighs] How many squares compose a cube pattern? A cross (range of 4 squares + 1 square on both sides) is it suitable?

Rep. 6 squares; yes, the cross folds into cubes.

5. Red thread · Flash [SOL] How many faces, peaks and edges have a square-based pyramid?

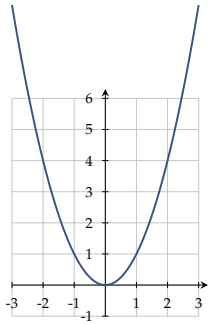
Rep. Five faces, five peaks, eight edges.

Form 23 Mon. 14 Dec. 2026 Graphical and numerical resolution: f $(x) = k$, $f(x) < k$

1. Warming · Flash [EQ] Solve $x^2 = 16$ and $x^2 = -4$.

Rep. $x = \pm 4$; no real solution ($x^2 \geqslant 0$).

2. Heart · Graph [REG] Curve of $f(x) = x^2$:



Graphically resolve $f(x) = 4$ then $f(x) < 4$.

Rep. $f(x) = 4 : x = -2$ or $x = 2$; $f(x) < 4 : x \in]-2;2[$.

Search the abscisses of the points of the curve below the right y = 4.

3. Heart + · Method [EQ] Solve $|x| < 3$ and $|x| = 5$.

Rep. $]-3;3[$; $x = -5$ or $x = 5$.

4. Back · Trap [SUB] Calculate $-x^2$ then $(-x)^2$ for $x = -3$.

Rep. $-(-3)^2 = -9$ and $(-(-3))^2 = 9$.

5. Red thread · Method [DEM] In a triangle ABC, M and N are the midpoints of $[AB]$ and $[AC]$. That What property is quoted?

Rep. $(MN) \parallel (BC)$ and $MN = \frac{1}{2}BC$: theorem on the right of the circles.